Type MS · Calculate BMI and place it in a weight category

Answer 1 of 6 · Question 3 part b.i · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

Number				
3	(a) (i)	3 320 000;		1
	(ii)	6.00239; x 10^5;	Allow 6.0024	2
	(iii)	pancreas;		1
	(b) (i)	120 ÷ 1.95^2/3.8029; 31.6;	Allow figure not rounded - 31.5581 Allow 31.56/32	2
	(ii)	obese;	ecf from previous question	1
	(c) (i)	• as BMI increases the risk of diabetes increases; • between a BMI of 24-35 the risk of diabetes increases more rapidly/BMI less than 25 the risk increases less rapidly;	Allow any range values between 24-35	2
	(ii)	Any two from the following: • balanced diet; • eat less fat; • eat less sugar	Ignore healthy diet Allow reduce alcohol/carbohydrate consumption for mp3	2

Total marks 11

Type MS · Calculate BMI and place it in a weight category

Answer 1 of 6 · Question 3 part b.i · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate BMI and place it in a weight category

Answer 2 of 6 · Question 5 part b.i · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

5	(a)	(i)	175.9 + 178.5 + 166.3 + 167.6 ÷ 4;		1
		Clip with table	172.1;	Full marks for correct final answer Reject 172	1
		(ii)	Height increases with age (for males / females / both); Males are taller/females are shorter (at all ages); Females increase in height greater compared with males between the ages of 12 and 18; Greatest height increase between ages 12 to 14;		Max 2
		(iii)			1
			Use more students in each age group; Use a greater age range of students; Use ages with 1 year gap/13,15,17;	Allow other valid improvements Ignore repeats	1
	(b)	(i)	71 ÷ 1.46^2;		1
		Clip with table	33/33.3;	Full marks for final correct answer One mark for correct calculation with incorrect figures	1
		(ii)	student Y is obese;	ECF from b(i)	1
		CLIP (b)(i) with (b)(ii)	higher risk of diabetes/heart disease/arthritis/cancers;	Allow other medical condition linked to obesity	1

Total question 5 = 10 marks

Type MS · Calculate BMI and place it in a weight category

Answer 2 of 6 · Question 5 part b.i · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate BMI and place it in a weight category

Answer 3 of 6 · Question 3 part b.ii · 1 mark

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

Number				
3	(a) (i)	3 320 000;		1
	(ii)	6.00239; x 10^5;	Allow 6.0024	2
	(iii)	pancreas;		1
	(b) (i)	120 ÷ 1.95^2/3.8029; 31.6;	Allow figure not rounded - 31.5581 Allow 31.56/32	2
	(ii)	obese;	ecf from previous question	1
	(c) (i)	• as BMI increases the risk of diabetes increases; • between a BMI of 24-35 the risk of diabetes increases more rapidly/BMI less than 25 the risk increases less rapidly;	Allow any range values between 24-35	2
	(ii)	Any two from the following: • balanced diet; • eat less fat; • eat less sugar	Ignore healthy diet Allow reduce alcohol/carbohydrate consumption for mp3	2

Total marks 11

Type MS · Calculate BMI and place it in a weight category

Answer 3 of 6 · Question 3 part b.ii · 1 mark

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate BMI and place it in a weight category

Answer 4 of 6 · Question 7 part b.ii · 3 marks

2024 JUN · Paper 02 · find part b.ii in this mark scheme extract

7 (a)	Any four from: • less blood flow/blood flow slows/restricted; • to heart muscle/tissue/cells; • less oxygen/glucose; • less aerobic respiration; • heart muscle/tissue/cells die/angina/heart attack;		(4)
(b) (i)	• (women 29 ÷ 100 (x 33 500 000) 9 715 000; • (men 26 ÷ 100 (x 33 000 000) 8 580 000; • (9 715 000 – 8 580 000) 1 135 000/1.135 million/1.135×10^6;	ecf for mp3	(3)
(ii)	• calculate BMI; • mass÷(height)^2; • compare result to BMI chart/ obese if 30 or more;	Accept • waist: hip ratio; • waist÷hip • Compare result to hip:waist ratio chart/	(3)
(c)	• beta blockers bind/ attach to receptors (on cardiac muscle); • prevent neurotransmitters from binding/attaching (to receptors); • reduce heart rate/reduced blood pressure;	Accept beta blockers fits into receptors accept neurotransmitters do not fit into receptors	(3)

Total for Question 7 = 13 marks

Type MS · Calculate BMI and place it in a weight category

Answer 5 of 6 · Question 4 part b.i · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

4 (a)	(i)	• glucose (units); • linked/joined together;		2
	(ii)	insulin;		1
	(iii)	glucagon:		1
(b)	(i)	• $67.0 \div 1.69^2$; • 23.5/23.4/23.46/23.458;		2
	(ii)	healthy weight *less than 18.5 underweight* *18.5 to 24.9 healthy weight* *25 to 29.9 overweight* *30 to 34.9 obese* *35 to 39.9 severely obese* *more than 39.9 morbidly obese*	ECF value in (b)(i)	1
(c)		Any two from: • fatty deposits/plaques/atheroma; • in coronary arteries; • reduced blood flow to heart muscles/tissue; • lack of oxygen/glucose to heart muscle/tissue;		2

Total Marks 9

Type MS · Calculate BMI and place it in a weight category

Answer 5 of 6 · Question 4 part b.i · 2 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate BMI and place it in a weight category

Answer 6 of 6 · Question 4 part b.ii · 1 mark

2024 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

4 (a)	(i)	• glucose (units); • linked/joined together;		2
	(ii)	insulin;		1
	(iii)	glucagon:		1
(b)	(i)	• $67.0 \div 1.69^2$; • 23.5/23.4/23.46/23.458;		2
	(ii)	healthy weight *less than 18.5 underweight* *18.5 to 24.9 healthy weight* *25 to 29.9 overweight* *30 to 34.9 obese* *35 to 39.9 severely obese* *more than 39.9 morbidly obese*	ECF value in (b)(i)	1
(c)		Any two from: • fatty deposits/plaques/atheroma; • in coronary arteries; • reduced blood flow to heart muscles/tissue; • lack of oxygen/glucose to heart muscle/tissue;		2

Total Marks 9

Type MS · Calculate BMI and place it in a weight category

Answer 6 of 6 · Question 4 part b.ii · 1 mark

2024 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks